

Hydrogen Atom :

For electron (e) and proton (p)

$$\begin{aligned} & [(\hat{x}_p)_i, (\hat{p}_p)_j] = i\hbar \delta_{ij} \\ & \& [(\hat{x}_e)_i, (\hat{p}_e)_j] = i\hbar \delta_{ij} \end{aligned} \quad i, j = 1, 2, 3 \quad (3D)$$

$$\psi = \psi(x_p, x_e)$$

$$\hat{H} = \frac{\hat{p}_e^2}{2m_e} + \frac{\hat{p}_p^2}{2m_p} + V(|x_e - x_p|)$$

$\Rightarrow$ COM variables : (1st pair of canonical eqs)

$$\hat{P} = \hat{p}_e + \hat{p}_p \quad ; \quad \hat{X} = \frac{m_e \hat{x}_e + m_p \hat{x}_p}{m_e + m_p}$$

$$[(\hat{X})_i, (\hat{P})_j] = i\hbar \delta_{ij}$$

$\Rightarrow$ 2nd pair of canonical eqs :

$$\hat{x} = \hat{x}_e - \hat{x}_p \quad (\text{relative pos})$$

$$\hat{p} = \alpha \hat{p}_e - \beta \hat{p}_p$$

$$\text{where, } \alpha = \frac{\mu}{m_e}, \quad \beta = \frac{\mu}{m_p}$$

$$\hat{p} = \mu \left(\hat{p}_e, \hat{p}_p \right) : \hat{x} = \hat{x}_e - \hat{x}_p$$

$$\hat{H} = \frac{\hat{p}^2}{2M} + \frac{\hat{p}^2}{2\mu} + V(\hat{r})$$

=> Solving Schrodinger eq (time-indp):

$$\Psi(X, r) = \Psi_{cm}(X) \cdot \Psi_{rel}(r)$$

Plugging into : $H\Psi = E\Psi$

$$\underbrace{\frac{1}{\Psi_{cm}(X)} \left[\frac{\hat{p}^2}{2M} \Psi_{cm}(X) \right]}_{f(X)} + \underbrace{\frac{1}{\Psi_{rel}(r)} \left[\frac{\hat{p}^2}{2\mu} + V(\hat{r}) \right] \Psi_{rel}(r)}_{f(r)} = \underbrace{E}_{\text{constant}}$$

Thus,

$$\frac{\hat{p}^2}{2M} \Psi_{cm}(X) = E_{cm} \Psi_{cm}(X)$$

$$\left[\frac{\hat{p}^2}{2\mu} + V(\hat{r}) \right] \Psi_{rel}(r) = E_{rel} \Psi_{rel}(r)$$

$$E = E_{rel} + E_{cm}$$

Spectrum :

- Central potential : $V(r) = -\frac{Ze^2}{r}$

- Fine structure constant : $\alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137}$

- Bohr's Radius : $a_0 = \frac{\hbar^2}{me^2} \approx 53 \text{ pm}$

$$\hookrightarrow K \cdot E = P \cdot E$$

$$\frac{\hbar^2}{me a_0^2} = \frac{e^2}{a_0}$$

- Energy scale : $\frac{e^2}{a_0} \approx 27.2 \text{ eV}$

Radial Schrodinger Eq for Bound states $E < 0$:

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2mr^2} - \frac{Ze^2}{r} \right) u = Eu$$

For a unit-free coord ρ : $\rho \equiv \frac{a_0}{2Z} \rho$

$$\Rightarrow \left(-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} - \frac{1}{\rho} \right) u = \frac{E}{\left(\frac{2Ze^2}{a_0} \right)} u$$

$$\text{let } \kappa^2 = - \frac{E_0}{\left(\frac{2Ze^2}{a_0} \right)} > 0$$

$$\left(-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} - \frac{1}{\rho} \right) u = -\kappa^2 u$$

$$\Rightarrow \text{for } n \rightarrow \infty : u \sim e^{-\kappa n}$$

Since κn is unit free assign it as: $\boxed{\rho = \kappa n}$

$$\Rightarrow \left(\frac{-d^2}{d\rho^2} + \frac{\ell(\ell+1)}{\rho^2} - \frac{1}{\kappa\rho} \right) u = -u$$

$$\Rightarrow \text{for } \rho \rightarrow 0 : u \sim \rho^{\ell+1}$$

Thus we can get:

$$u(\rho) = \rho^{\ell+1} \cdot \underset{\substack{? \\ ? \\ ?}}{w(\rho)} \cdot e^{-\rho}$$

Principle Quantum No. (n):

$$\boxed{n \equiv N + \ell + 1 = \frac{1}{2\kappa}}$$

Energy:

$$\boxed{E = -\frac{Z^2 e^2}{2a_0} \times \frac{1}{n^2}}$$

Number of States

n value	ℓ values	m values	total states
$n=1$	$\ell=0$	$m=0$	1 state
$n=2$	$\ell=0$	$m=0$	1
	$\ell=1$	$m=-1, 0, 1$	+ 3 = 4 states
$n=3$	$\ell=0$	$m=0$	1
	$\ell=1$	$m=-1, 0, 1$	+ 3 = 4 states

$\ell=1$	$m = -1, 0, 1$	+3
$\ell=2$	$m = -2, \dots, 2$	+5
		= 9 states

Total no. of states: $\sum_{\ell=0}^{n-1} (2\ell+1) = \underline{\underline{n^2}}$

"The final Wave Equation":

$$\Psi_{nlm}(r, \theta, \phi) = N \cdot \left(\frac{r}{a_0}\right)^\ell \left(\begin{array}{c} \text{polyn. in } r/a_0 \text{ of} \\ \text{deg. } N=n-(\ell+1) \end{array} \right) e^{-\frac{Zr}{na_0}} Y_{\ell m}(\theta, \phi)$$

For: $\Psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} \cdot e^{-r/a_0}$

$$\left. \begin{array}{l} n-(\ell+1)=0 \quad (\text{poly}) \\ \ell=0 \rightarrow (r/a_0)^\ell \\ m=0 \rightarrow (Y_{\ell m}) \end{array} \right\} \rightarrow n=1$$

Runge-Lenz vector

Hamiltonian: $H = \frac{p^2}{2m} - \frac{e^2}{r}$

Natural Scale:

• Length: Bohr radius $\rightarrow a_0 = \frac{\hbar^2}{me^2} \simeq 0.59 \text{ \AA}$

• Energy: $E_0 = \frac{e^2}{a_0} = \alpha^2 (mc^2)$

Recall: $\alpha = \frac{e^2}{\hbar c} \sim \frac{1}{137}$

Ground State: $\left(\hat{p}_\kappa - i\beta \frac{\hat{\chi}_\kappa}{r} \right) |\psi_0\rangle = 0$

